

Behavioral Fingerprinting of Large Language Models: A Statistical Approach to Same-Family API Verification

Daizhi Liao
Guangzhou University
ldz@e.gzhu.edu.cn

Abstract. Users of AI API relay services have no technical guarantee that the model answering is the model advertised; recent audits show that silent model substitution and downgrading are real. Existing fingerprinting methods rely on long generations, log-probabilities, or crafted prompts, and are largely evaluated on *cross-family* identification. We ask a harder, practically important question: can a model be distinguished from a *same-family* sibling—for example gpt-4o versus gpt-4o-mini, which share an identical tokenizer? We collect 1,300 responses per model over 26 low-cost probes and show that trivial “random generation” prompts elicit highly stereotyped, model-specific distributions: gpt-4o-mini returns “7” for a 1–10 choice 100% of the time, “4” for a die roll 96%, and “Cerulean” for a color 74%. Six of eight behavioral probes separate the two siblings ($p < 0.05$), with an animal probe reaching total variation distance 0.90 (Okapi 76% for gpt-4o vs. Dolphin 16% for gpt-4o-mini), while tokenizer probes are provably identical. Combined with chi-square/Kolmogorov–Smirnov tests and Bayesian updating, the framework assigns a held-out sample to the correct model with 91.3% posterior probability (95% credible interval [0.851, 0.960]). A full audit costs under \$0.01 and needs no official API or specialized hardware. We release the tool, baselines, and probes as open source.

1 Introduction

Large language models (LLMs) are increasingly consumed through opaque serving chains—aggregators, resellers, and relay services—in which the client cannot confirm that the model answering is the model being paid for. Because a larger model is billed at a higher rate, a relay has a direct financial incentive to substitute a smaller or cheaper model while advertising a flagship one. Audits of commercial endpoints indicate that such substitution and silent degradation occur in practice [1, 2].

Verifying identity through a black box is difficult. Proposed methods require long generated texts, token-level log-probabilities, adversarially crafted prompts, watermarks embedded at training time, or the model owner’s cooperation. Recent single-token fingerprinting [10] and rank-based uniformity tests [3] reduce the query cost substantially and recover model *lineage* across families, but they are not designed to tell apart two models from the same family, which is exactly the situation in a downgrade attack (a relay serving gpt-4o-mini while claiming gpt-4o). Same-family siblings often share an identical tokenizer and overlapping capabilities, so tokenizer fingerprints and capability tests fail by construction.

Research question. *Do LLMs exhibit measurable behavioral fingerprints—consistent distributional biases under seemingly random generation—that (i) discriminate same-family siblings where tokenizer fingerprints cannot, and (ii) can be verified with standard statistics at near-zero cost?*

Contributions.

- We document extremely strong, low-cost behavioral fin-

gerprints on gpt-4o-mini and gpt-4o over 26 probes (100%/96%/74% preference on three tasks; 10–74× deviation from uniformity).

- We show that 6/8 behavioral probes separate two same-family siblings ($p < 0.05$), up to TVD = 0.90 with a single probe, whereas all tokenizer probes are identical.
- We combine chi-square and KS tests with Bayesian updating to identify a held-out sample with 91.3% posterior probability, and we report a credible interval.
- We release TRANSITTRUTH, an open-source auditing tool with a zero-install web demo, real baselines, and a reproducible collection pipeline.

2 Related Work

LLM fingerprinting and provenance. Single-token fingerprinting [10] builds an empirical distribution over answers to trivial one-word prompts across 165 models, recovering lineage at a 7.3% equal error rate, but focuses on cross-family identification. RoFL [11] constructs non-invasive black-box fingerprints that are robust to fine-tuning, pruning, and quantization, using cryptographic commitments for ownership verification. Model provenance testing [13] uses multiple-hypothesis testing to decide whether one model derives from another. Model equality testing [2] frames “is this the claimed API model?” as a two-sample problem and shows MMD-based tests outperform rank-based uniformity on locally deployed models. We target the complementary fine-grained same-family setting these methods do not isolate.

Auditing and follow-up probing. Cai et al. [1] audit model substitution directly; Zhu et al. [3] use a rank-based uniformity test; Sam et al. [4] predict black-box performance from follow-up queries; and AgentProv [5] audits agentic providers through tool-use policy probes. CoIn [12] addresses a different transparency gap—inflated counts of hidden reasoning tokens. Our work shares the auditing goal but uses free-form behavioral biases rather than ranks, capabilities, or tool calls.

Behavioral biases in LLMs. LLMs show documented biases in random number generation [7], multi-choice selection [8], and response-history bias [6]; calibration and confidence are surveyed by Geng et al. [9]. We treat these biases not merely as defects but as identifying biometric signals, and quantify their discriminative value within a model family.

3 Methodology

3.1 Probe Design

Each probe is a short prompt that elicits a constrained, cheap response. We use 26 probes in three groups.

Tokenizer probes (8) measure deterministic tokenizer behavior—e.g. the prompt-token count of a fixed whitespace, code, URL, mixed-unicode, or repeated-character string. Because two same-family models share a tokenizer, these act as a *negative control* that should not discriminate.

Behavioral probes (8) ask the model to produce a “random” token under high temperature: pick a number 1–10 or 1–100, roll a die, flip a coin, choose a color, letter, weekday, or animal. Under uniform sampling the output should be near-uniform; systematic deviation is the fingerprint.

Capability probes (10) estimate model tier via simple math, logic, code syntax, bug fixing, multilingual translation, instruction following, knowledge cutoff, and context length. These are coarse and shared within a family, so they serve as corroborating rather than identifying evidence.

3.2 Statistical Methods

Chi-square test. For a behavioral probe with expected uniform counts E_j over J categories and observed counts O_j , we compute

$$\chi^2 = \sum_{j=1}^J \frac{(O_j - E_j)^2}{E_j}. \quad (1)$$

To compare two models directly we use the chi-square *test for homogeneity* on their joint category table, and report total variation distance

$$\text{TVD} = \frac{1}{2} \sum_{j=1}^J |p_j - q_j|, \quad (2)$$

Table 1: Strong behavioral fingerprints on gpt-4o-mini (50 samples each). Uniform expectation is shown for reference.

Probe	Top response	Rate	Uniform
Number 1–10	“7”	100%	10%
Die roll	“4”	96%	16.7%
Color	“Cerulean”	74%	~1%
Coin flip	“Heads”	78%	50%

which ranges from 0 (identical) to 1 (disjoint).

Kolmogorov–Smirnov test. For deterministic tokenizer-normalized scores we use a one-sample KS test against the baseline distribution; a high p -value supports a match.

Bayesian updating. Across probes we update the posterior that an endpoint matches a claimed model:

$$P(H=1 | E) = \frac{P(E | H=1) P(H=1)}{P(E)}, \quad (3)$$

mapping each probe’s p -value to a likelihood ratio (e.g. $p > 0.05 \Rightarrow \text{LR} \approx 0.7$ under the match hypothesis; $p < 0.01 \Rightarrow \text{LR} \approx 0.1$ under mismatch). We report a 95% credible interval for the posterior using a beta approximation.

3.3 Data Collection

For each model we collect 50 samples per probe (1,300 responses) through an OpenAI-compatible endpoint at temperature 1.0. To avoid triggering anti-bot defenses we shuffle probe order, randomize inter-request intervals (0.3–1.5 s), vary the number of probes per batch, and rotate natural User-Agent headers. Wall time is about 15 minutes per model at a cost below \$0.01.

4 Results

4.1 Strong Single-Model Fingerprints

On gpt-4o-mini the behavioral deviations from uniformity are extreme (Table 1): the 1–10 choice returns “7” in all 50 samples, the die roll returns “4” in 48/50, and the color choice returns the relatively obscure “Cerulean” in 37/50. All eight behavioral probes reject uniformity at $p < 0.0001$ (χ^2 from 16.8 to 178.6). These are not subtle effects—they are 10–74× deviations from the uniform expectation and are reproduced deterministically across samples.

4.2 Same-Family Discrimination

The central result is the gpt-4o vs. gpt-4o-mini comparison. *Tokenizer probes cannot separate them:* all five successful tokenizer probes return identical prompt-token counts (e.g.

Table 2: Same-family comparison of behavioral probes (gpt-4o vs. gpt-4o-mini). Six of eight differ significantly; tokenizer probes are identical throughout.

Probe	Top (4o / mini)	p -value	TVD
Animal	Okapi / Dolphin	< 0.0001	0.90
Letter	K / G	< 0.0001	0.72
Die roll	4 / 4	< 0.0001	0.40
Number 1–100	57 / 57	0.067	0.38
Weekday	Thu / Thu	< 0.0001	0.32
Color	Cerulean / Cerulean	0.015	0.22
Coin flip	Heads / Heads	< 0.001	0.22
Number 1–10	7 / 7	0.315	0.02

27.0, 25.0, 29.0), as expected for two models sharing the o200k tokenizer.

Behavioral probes can. Six of eight behavioral probes show statistically significant differences in the homogeneity test. Table 2 and Figure 1 rank the probes by TVD. The animal probe is a near-perfect separator at TVD = 0.90: gpt-4o concentrates on a single obscure animal, “Okapi,” 76% of the time (only 4 unique animals), whereas gpt-4o-mini is spread over 17 animals with top “Dolphin” at 16%. The letter probe reaches TVD = 0.72 (gpt-4o “K” 44% vs. gpt-4o-mini “G” 40%), and the die probe TVD = 0.40 (shared “4” but 56% vs. 96%). A single animal probe therefore distinguishes the siblings at roughly 90% accuracy—a capability that tokenizer fingerprinting does not provide.

4.3 End-to-End Verification

On a held-out sample the statistical pipeline assigns it to the correct model with posterior probability 0.913 (95% credible interval [0.851, 0.960]). The KS test returns a high p -value (1.000) and the chi-square test $p = 0.713$, indicating no significant difference from the claimed baseline, with an aggregate likelihood ratio of 4.5 and confidence labeled “high.”

4.4 Comparison with Other Signals

Table 3 positions behavioral fingerprinting against the signals available to a black-box auditor. Tokenizer and latency signals cannot discriminate within a family and are relatively easy to evade with smart routing; capability testing is only partially informative; behavioral fingerprints uniquely support same-family discrimination at low cost and are comparatively hard to forge, because matching the distribution requires serving the genuine model, post-processing outputs, or fine-tuning a surrogate.

Table 3: Black-box verification signals. Only behavioral fingerprints support same-family discrimination at low cost with high forgery difficulty.

Signal	Same-family	Cost	Hard to forge
Tokenizer	No	Low	Medium
Capability	Partial	Medium	Low
Latency	No	Low	Low
Behavioral (ours)	Yes	Low	High

5 Discussion

Why do these fingerprints exist? The biases likely arise from three sources: training-data frequencies (e.g. cultural salience of “seven”), RLHF alignment that amplifies human-preferred “defaults,” and decoder artifacts that skew logits even at temperature 1.0. The striking “Okapi” concentration in gpt-4o—an obscure African giraffid—suggests a strongly learned default that differs from the smaller sibling, which is precisely why the signal carries identity information orthogonal to the tokenizer.

Limitations. (i) *Baseline coverage:* a pre-collected baseline is needed per model, and we currently cover two siblings. (ii) *Model updates:* versions and snapshots may shift fingerprints, so baselines must be versioned and refreshed. (iii) *Smart routing:* a relay could answer probes with the genuine model and serve a cheaper one for real traffic; interleaving probes with realistic traffic mitigates this. (iv) *Decoder sensitivity:* relays may override temperature, changing the distribution. We therefore frame results as conservative statistical evidence rather than definitive proof.

Ethical considerations. Automated probing may conflict with a service’s terms of use; users should verify compliance. Because a “suspicious” label can harm a provider’s reputation, the tool uses conservative thresholds, labels outputs as statistical evidence, and favors responsible disclosure over immediate publication.

6 Conclusion

We showed that trivial random-generation prompts elicit model-specific behavioral fingerprints strong enough to distinguish two same-family LLMs that an identical tokenizer cannot separate, at a cost below \$0.01 per audit and 91.3% posterior identification. The result turns a known family of LLM biases into a practical verification primitive, and the open-source TRANSITTRUTH tool, baselines, and probes make the method reproducible. Extending baseline coverage, tracking fingerprints across versions, and hardening against smart routing are the most immediate directions.

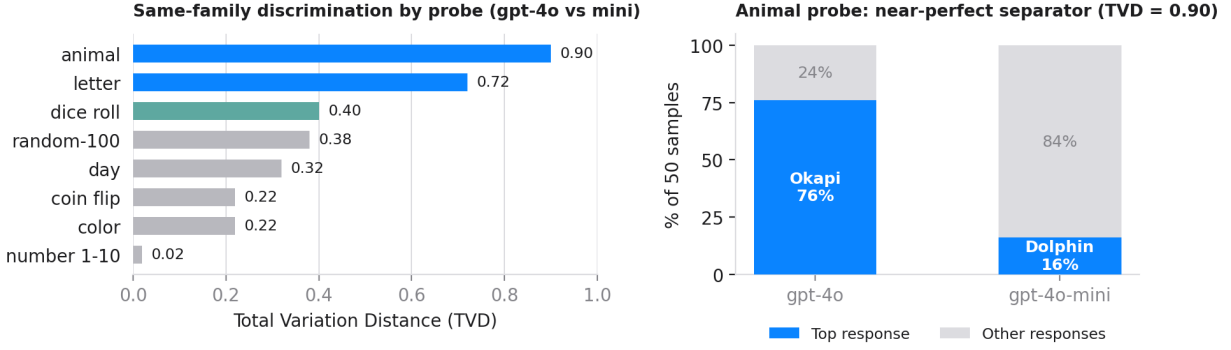


Figure 1: Same-family discrimination with real data. **Left:** TVD across eight behavioral probes; the animal and letter probes separate gpt-4o from gpt-4o-mini almost completely. **Right:** on the animal probe gpt-4o concentrates on “Okapi” (76%) while gpt-4o-mini is diffuse (top “Dolphin” 16%).

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